

DYNAMICAL SYSTEMS - MA 3081

Retake exam - solutions

Answers always require detailed justification, e.g. in the form of deductions, calculations, or clear references to results from the lecture. You can answer the questions in any order, but always indicate which question you are working on. Answers to one question should not be split into multiple pieces, but written coherently. Do not use the color red.

Problem 1. For $p \in \mathbb{R}$, consider the system

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} -(1+p)x + x^2y + 1 \\ px - x^2y \end{pmatrix}.$$

- (a) Find the equilibria of the system as a function of the parameter p .
- (b) Determine the stability of the equilibria when they are hyperbolic.
- (c) Show that a Hopf bifurcation occurs for a certain value $p = p_0$.

Solution:

- (a) We need to solve

$$\begin{aligned} 1 - (p+1)x + x^2y &= 0, \\ px - x^2y &= 0. \end{aligned}$$

The second equation is satisfied for $x = 0$ or $p = xy$. If $x = 0$, the first equation becomes $1 = 0$, which clearly has no solution. Thus, we must have $p = xy$. Inserting this into the first equation we get

$$1 - (p+1)x + px = 1 - x = 0,$$

i.e. we have $x = 1$ and therefore $y = p$. Hence the only equilibrium of the system is given by $(x, y) = (1, p)$ for each $p \in \mathbb{R}$. [2 points]

- (b) We first compute the derivative of the vector field $f(x, y) = \begin{pmatrix} 1 - (p+1)x + x^2y \\ px - x^2y \end{pmatrix}$.

It is given by

$$Df(x, y) = \begin{pmatrix} -(p+1) + 2xy & x^2 \\ p - 2xy & -x^2 \end{pmatrix},$$

so the evaluation at the equilibrium $(1, p)$ yields

$$Df(1, p) = \begin{pmatrix} p-1 & 1 \\ -p & -1 \end{pmatrix}$$

This matrix has eigenvalues $\lambda_{1,2} = \frac{1}{2} \left((p-2) \pm \sqrt{p(p-4)} \right)$.

If $0 < p < 4$, the real part of those eigenvalues is $\frac{p-2}{2}$, so the equilibrium is hyperbolic as long as $p \neq 2$. Especially the equilibrium of the linearized system is stable, if $p < 2$ and unstable if $p > 2$.

If $p > 4$, both eigenvalues are real. Since $\det(Df(1, p)) = \lambda_1 \lambda_2 = 1$, they have the same sign and are always non-zero, i.e. hyperbolic. Furthermore $\text{tr}(Df(1, p)) = p - 2 = \lambda_1 + \lambda_2$, so if $p < 2$, the equilibrium of the linearized system is stable and if $p > 2$ it is unstable.

Consequently, for all $p \neq 2$ the equilibrium $(p, 1)$ of the nonlinear system is hyperbolic and by Theorem 1.33 (Hartman Grobman) it is stable if $p < 2$ and unstable if $p > 2$.

[5 points]

- (c) At $p = 2$ hyperbolicity is lost. For $p \approx 2$ we have a pair of complex conjugate eigenvalues $\lambda_{1,2} = \frac{p-2}{2} \pm i\sqrt{4 - (\frac{p-2}{2})^2}$. Furthermore, the real part $\mu(p) = \frac{p-2}{2}$ satisfies $\mu'(2) = \frac{1}{2} \neq 0$ and the imaginary part $\omega(p) = \sqrt{4 - (\frac{p-2}{2})^2}$ satisfies $\omega(2) = 2 \neq 0$. Thus by Theorem 4.5 a Hopf bifurcation happens at $p_H = 2$.

[1 point]

Problem 2. Prove that topological transitivity is preserved under topological conjugacy. More precisely, if two dynamical systems are topologically conjugate, then either both of them are topologically transitive or both of them are not topologically transitive.

Solution: Let $f : X \rightarrow X$ and $g : Y \rightarrow Y$ be two dynamical systems and $h : X \rightarrow Y$ a conjugacy between them. Assume $O \subset X$ is an orbit of f which is dense in X . We claim that then $h(O)$ is dense in Y . Indeed, let $y \in Y$ be arbitrary and V a neighborhood of y in Y . Assume that $h(O) \cap V = \emptyset$. Then $O \cap h^{-1}(V)$ is also empty, contradicting the fact that O is dense in X .

[4 points]

Problem 3. Consider the space $\Sigma = \{(s_k)_{k \in \mathbb{Z}} : s_k \in \{0, 1\}\}$ of bi-infinite sequences of two symbols and the shift map $\sigma : \Sigma \rightarrow \Sigma$, $(\sigma(s))_k = s_{k+1}$. Show that σ has infinitely many homoclinic orbits.

Solution: For $K \in \mathbb{N}$ define

$$s^K = (s_k^K)_k := \begin{cases} 0, & k < -K, \\ 1, & -K \leq k \leq K, \\ 0, & k > K. \end{cases}$$

This orbit is homoclinic to the fixed point $s = 0$ for every $k \in \mathbb{N}$, thus there are infinitely many homoclinic orbits.

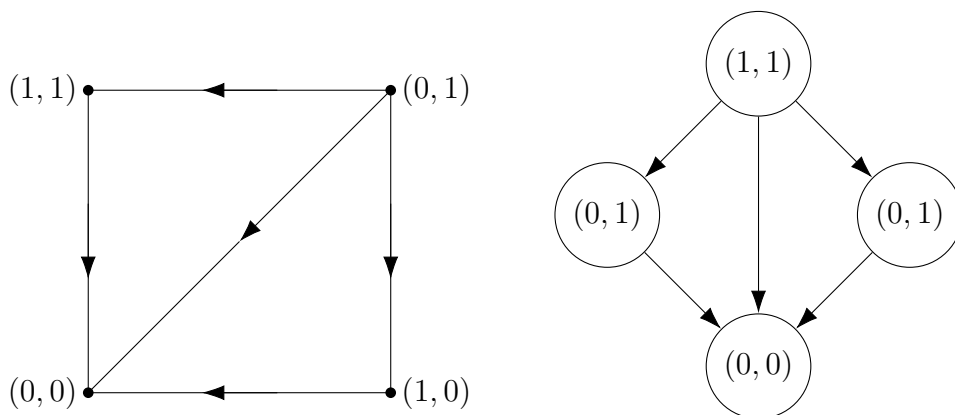
[4 points]

Problem 4. On the unit square $[0, 1]^2$, consider the flow induced by the differential equations

$$x' = x(x - 1), \quad y' = y(y - 1).$$

Find a Morse decomposition with at least three Morse sets and draw the corresponding diagram.

Solution: The system has four equilibria: $(0,0)$, $(0,1)$, $(1,0)$ and $(1,1)$. Here is the phase diagram and the corresponding Morse diagram where every equilibrium yields one Morse set: [4 points]



Problem 5. Consider the map $\varphi(x) = 2x + b \bmod 1$ on $[0,1]$, where $b \in \mathbb{R}$. Show that Lebesgue measure is an ergodic measure for φ . *Hint: Show first that if $f, g: \mathcal{X} \rightarrow \mathcal{X}$ are two maps for which some measure μ is invariant, then μ is also invariant for the map $h := g \circ f$. Then use the example from the lecture.*

Solution: Let B be a measurable set. Since μ is invariant for f and for g , we have

$$\mu(f^{-1}(g^{-1}(B))) = \mu(f^{-1}(B)) = \mu(B),$$

so that μ is invariant for $h = f \circ g$.

We now write $\varphi = f \circ g$ with $g(x) = 2x \bmod 1$ and $f(x) = x + b \bmod 1$. Since Lebesgue measure is invariant for both f and g , it is also invariant for φ .

Now g is exact, i.e. $\lim_{k \rightarrow \infty} \mu(g^k(A)) = 1$ for any initial set with $\mu(A) > 0$. Since f does not change the Lebesgue measure of a set, we also have that $\lim_{k \rightarrow \infty} \mu(\varphi^k(A)) = 1$, i.e. φ is exact and thus ergodic since exactness implies ergodicity. [4 points]

Problem 6. For $p \in \mathbb{R}$, consider the system

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} p(y^2 + xy) \\ -x^2 - y \end{pmatrix}.$$

(a) Show that a center manifold at the origin is given by

$$W_{\text{loc}}^c = \{(x, y) \in \mathbb{R}^2 \mid y = -x^2 - 2px^4 + O(x^5)\}.$$

(b) Determine the stability of the origin in dependence of p . *Hint: Consider $p \neq 0$ first. For $p = 0$, draw the phase diagram and use this to determine the stability of the origin.*

Solution:

(a) Set $h(x) = -x^2 - 2px^4$. We aim at applying Theorem 4.10.

First, we find that the linearization at the origin is given by $\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$, so that $E^c = \text{span}\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right\}$ and $E^s = \text{span}\left\{\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}$. Furthermore, we clearly have $h(0) = h'(0) = 0$.

Finally, we check the invariance equation

$$(-2x - 8px^3)[p(-x^3 - 2px^5 + (-x^2 - 2px^4)^2)] - [x^2 + 2px^4 - x^2] = 2px^4 + \mathcal{O}(x^5) - 2px^4 = \mathcal{O}(x^5).$$

By Theorem 4.10 the center manifold is given as the graph of

$$H(x) = h(x) + \mathcal{O}(x^5) = x^2 - 2px^4 + \mathcal{O}(x^5).$$

[5 points]

(b) By Theorem 4.9 it is sufficient to consider the system

$$\begin{aligned} x' &= p(-x^3 + \mathcal{O}(x^4)), \\ y' &= -y. \end{aligned}$$

For the x -equation, in the vicinity of $x = 0$ the cubic term dominates. Hence, in case of $p < 0$, the vector field is positive for $x > 0$ and negative for $x < 0$. Thus the equilibrium $x = 0$ is unstable. In case of $p > 0$ the picture reverses and the vector field is negative for $x > 0$ and positive for $x < 0$. Thus the equilibrium $x = 0$ is stable then. Since the equilibrium $y = 0$ of the second equation is stable we can conclude: If $p < 0$, $(x, y) = (0, 0)$ is an unstable equilibrium, if $p > 0$, $(x, y) = (0, 0)$ is a stable equilibrium.

[2 points]

For $p = 0$ the equations read

$$\begin{aligned} x' &= 0, \\ y' &= -y - x^2. \end{aligned}$$

Hence, we have an equilibrium point whenever $y = -x^2$. Furthermore, $y' > 0$, if $y < -x^2$ and $y' < 0$, if $y > -x^2$. The phase portrait is depicted below. From that we can directly conclude that the origin is a stable equilibrium.

[1 point]

